
Sprint 4.MSD: **RESTful Services Model** **Implementation**

Sprint_4 •

Overview

REST Architectural Components Implementation:

- RESTful Services Implementations with @RestController annotation
- DTO Model Implementation with @Json declarative serialization mapping

Course Coverage:

- Theme_3 Courses: C 7, C 8
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RESTful Services Model Implementation

Spring RESTful Resource Services

- REST controller services to expose entities from *Repository* operations.
- REST implementation services will be annotated with
 - @RestController
 - @RequestMapping
- *Repository* will be @Autowired into RestControllers

Spring REST-RPC Services (Opt.)

- REST controller services to expose Business Workflow Service Operations
 - Business Workflow Services will be @Autowired into RestControllers
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DTO Model Implementation

DTO Classes

- Entities with optional serialization annotation for
 - JSON
 - @JsonRootName
 - @JsonIgnoreProperties
 - @JsonProperty

DTO Aggregates

- Root Entity relationships. e.g.: @OneToMany
 - @JsonManagedReference
 - Component Entities relationships: @OneToMany to avoid cycling:
 - @JsonBackReferences
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Deliverables

SpringBoot Projects

@SpringBootApplication main class

RestController definition services
classes annotated with

- **@RestController**
- **@RequestMapping**

@Entity classes annotated with
@Json for DTO
serialization/deserialization

SpringBoot configuration file:

- resource/**application.properties**

JUnit Test Classes

- REST tests with **SpringBoot**
RestTemplate HTTP Client
library.